6. TROUBLESHOOTING TOOLS

Course: Diploma in Cyber Security

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Introduction

Command-line tools are useful for checking basic network problems. I can use ipconfig, ping, tracert/traceroute and nslookup to check addressing, connectivity, routing path and DNS resolution.

Task 1: ipconfig

Run ipconfig and record the IPv4 address and default gateway. Take a screenshot of the output.

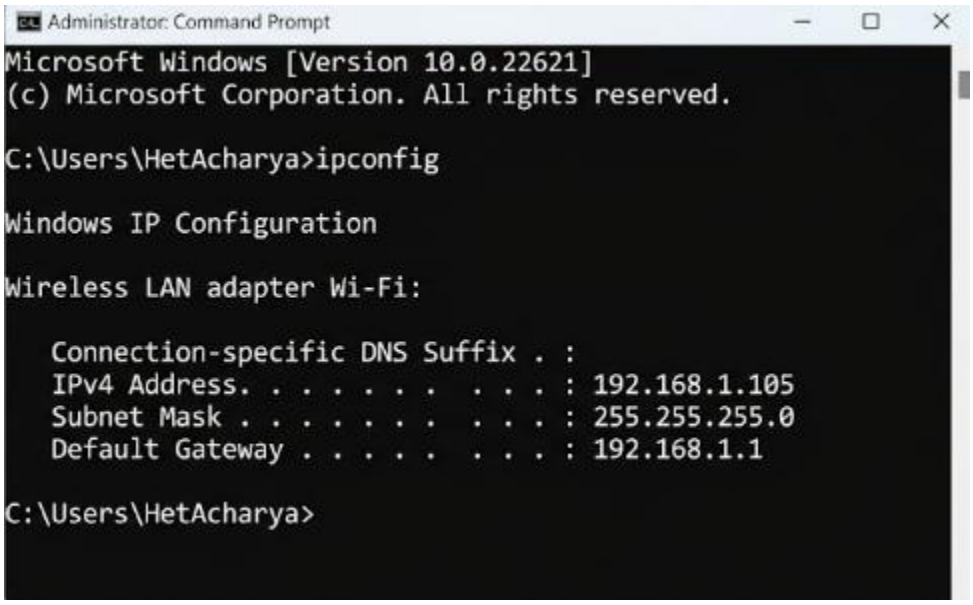


Figure 1: Windows Command Prompt ipconfig Output Displaying Local IPv4 Configuration

Task 2: ping

Run ping www.google.com and record the average response time shown by the command.

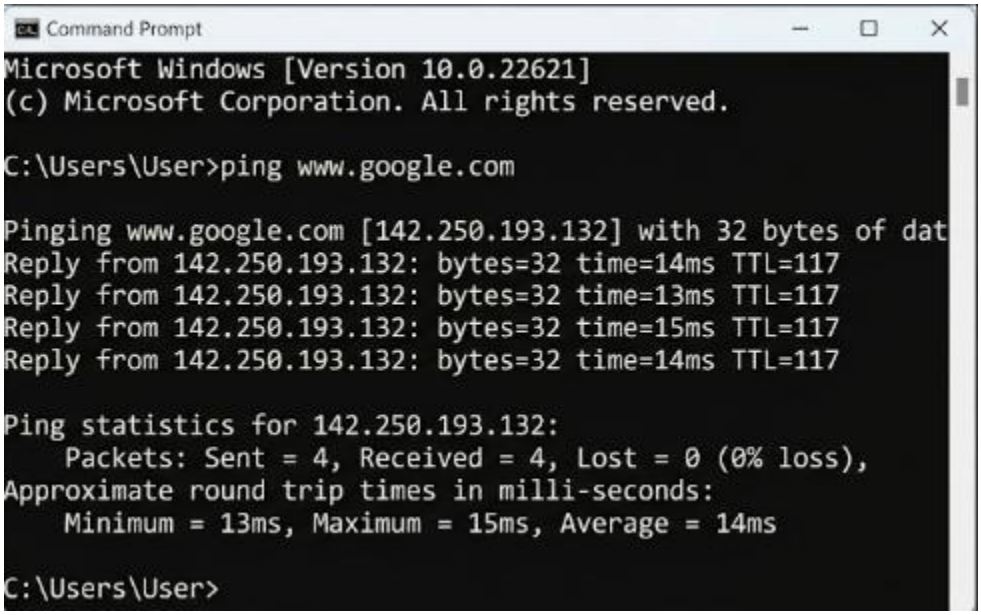


Figure 2: ICMP Echo Request Output to www.google.com Showing Latency Statistics

Task 3: tracert

Run tracert www.flipkart.com on Windows or traceroute on Linux/Mac and count the hops shown.

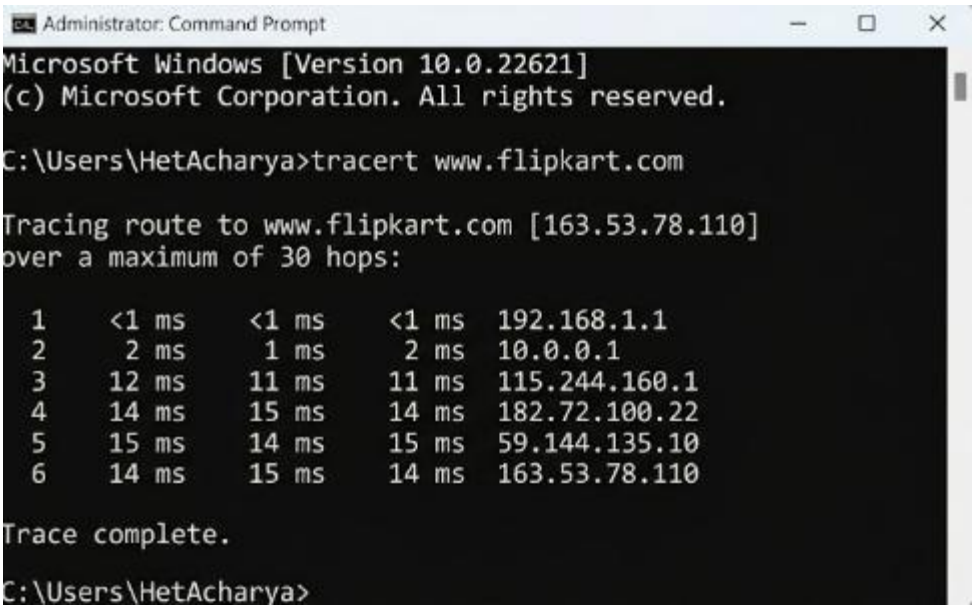


Figure 3: Layer 3 Path Tracing (tracert) to www.flipkart.com Detailing Hop Count

Task 4: nslookup

Run nslookup www.spotify.com. It helps check whether DNS can resolve the domain name to an IP address.

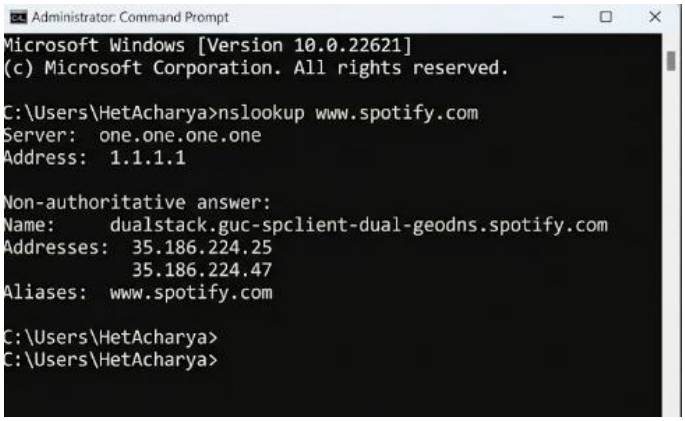
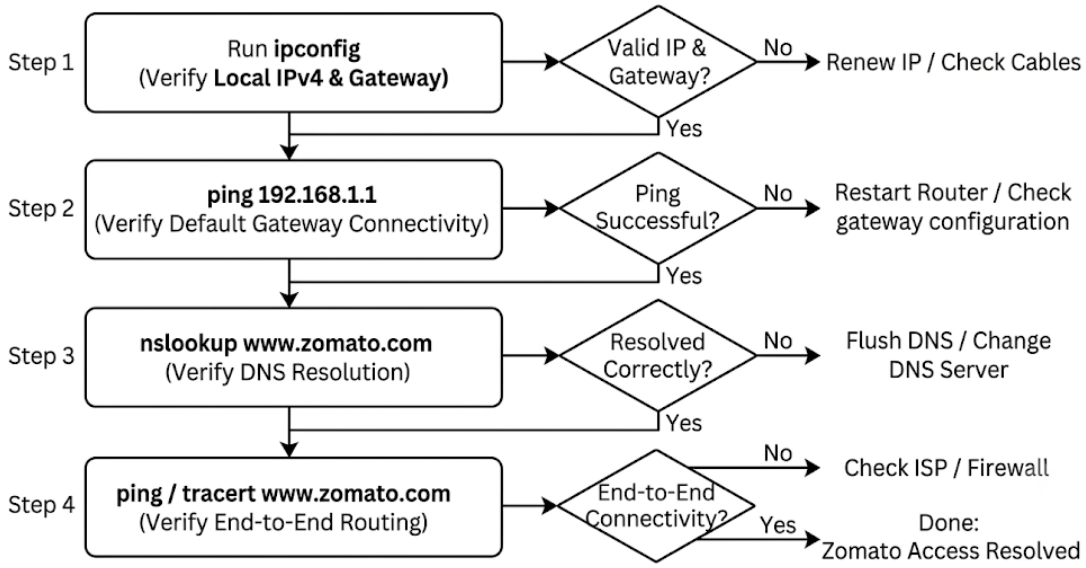


Figure 4: DNS Resolution Query for www.spotify.com via Default Nameserver

Task 5: Diagnose Zomato access

A simple sequence is ipconfig → ping the gateway/website → nslookup the domain. These steps check local configuration, connectivity and DNS.

Figure: Network Diagnostic Flow for Resolving Zomato Access



Conclusion

This topic helped me understand the basic concept and how it can be used in a real network. I also understood that practical results should be checked on the actual system or simulator instead of assuming the result.